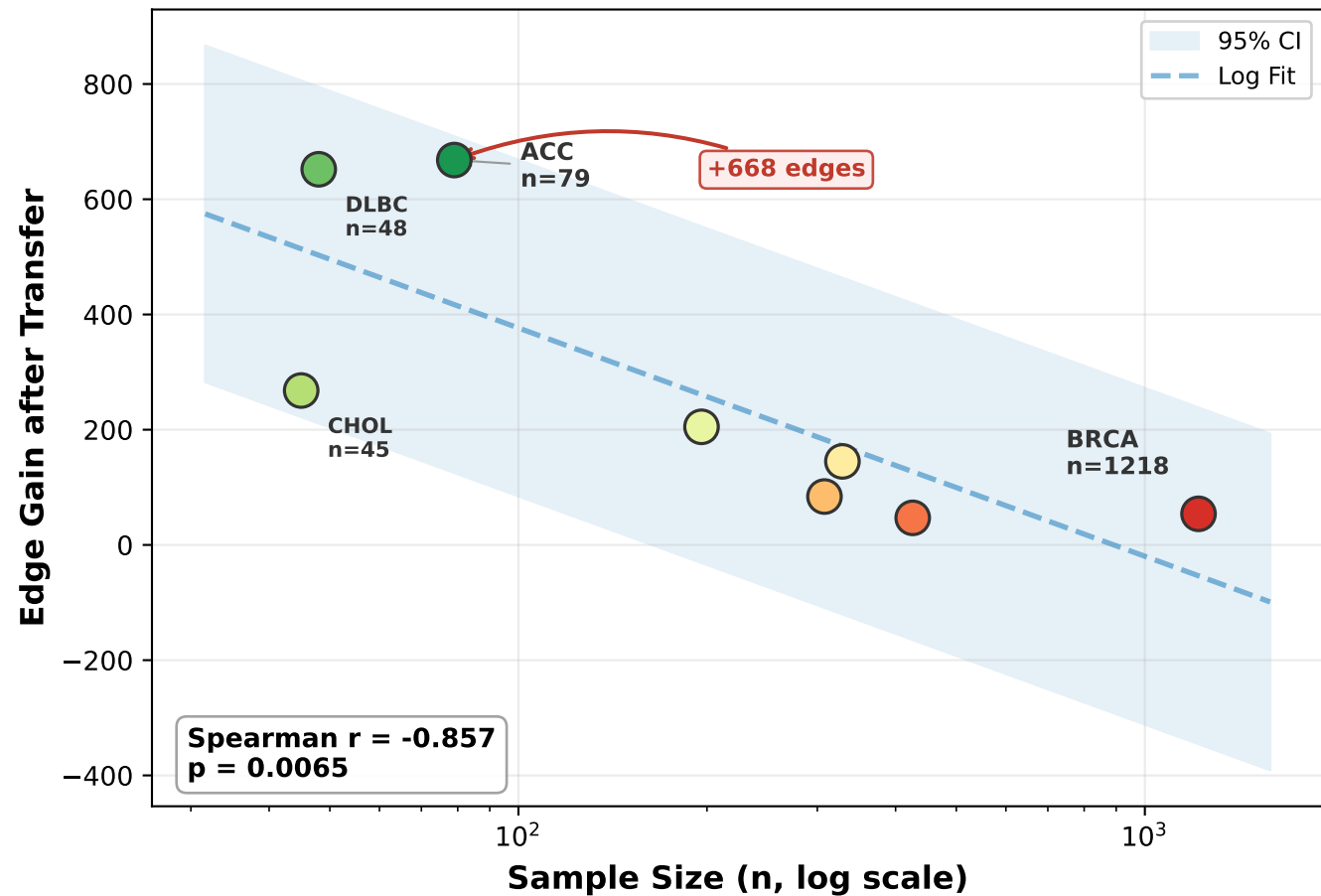
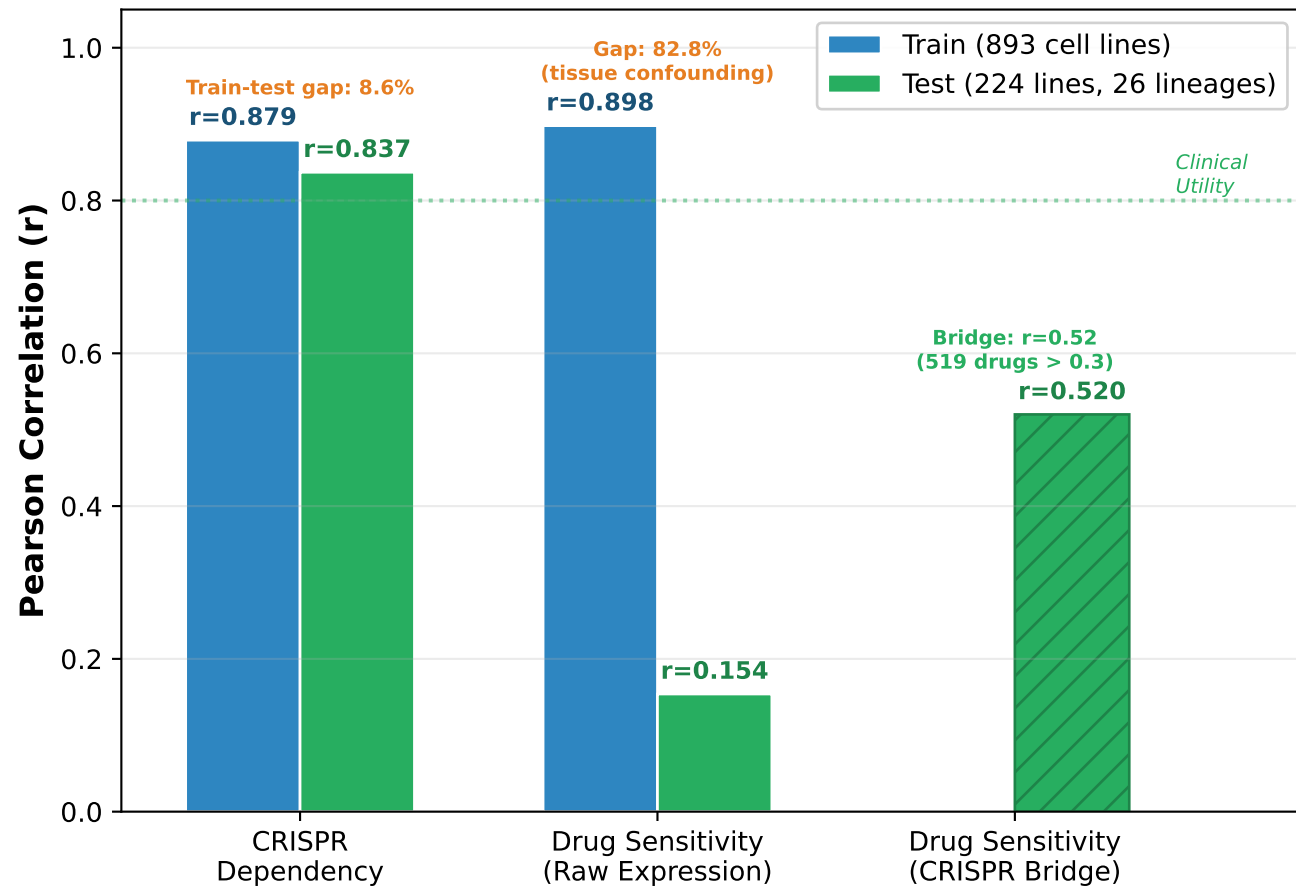


(a) Zero-Shot Transfer: Small-Sample Amplification



(b) Prospective Validation: 80/20 Stratified Split



DepMap pre-training + TCGA fine-tuning. CRISPR bridge uses causal features to bypass tissue-type confounding.